Online Course Analytics

A Power BI Dashboard for Content Strategy, Engagement & Market Positioning

Full dashboard view

An analysis of an online learning platform's course catalogue, built in Power BI to answer eight business questions about content strategy, learner engagement, and market positioning.

The Brief

A client wants to enter the online course market. Before committing to content production, they need to understand the existing landscape: which categories are crowded, what formats dominate, which skills are in demand, and what actually drives viewership.

The Data

The dataset covers online recorded courses across 11 categories and 46 sub-categories, with fields for course type, category, language, skills taught, duration, instructor, rating, and viewer counts.

Approach

Data was imported and shaped in Power Query, modelled in Power BI, and analysed through DAX measures. Two transformations were applied before analysis:

Duration normalisation — courses listed in months were converted at 60 hours per month; flexible self-paced courses were assigned a nominal 200 hours.

Duration binning — hours were grouped into 20-hour bands so averages rested on enough courses to be reliable.

Findings

FINDING 01

Course distribution across categories

The catalogue spans **11 categories and 46 sub-categories** with three course types: Course, Specialisation, and Guided Project. The standard Course format dominates in every single category. Specialisations and Guided Projects are a small minority throughout. Volume concentrates in **Business, Data Science, and Computer Science**, while **Personal Development and Math & Logic** have the thinnest catalogues.

11	46	3
Categories	Sub-categories	Course types

WHAT THIS MEANS

There's no category where the client would face real competition in Specialisations or Guided Projects. That's an open gap, especially in Data Science and Computer Science, where learners are most likely to want structured multi-course pathways.

FINDING 02

Average views by category and language

Category	Ranked_By_Average_Viewers
Data Science	**6327**
English	6364
Spanish	1740
Japanese	230
Information Technology	**4624**
English	4886
Japanese	31
Computer Science	**4353**
English	4422
Spanish	767
Personal Development	**3221**
English	3285
Spanish	252
Language Learning	**2844**
French	4468
English	2825
Total	**3212**

Categories ranked by average viewers, broken down by language

CATEGORY	AVG. VIEWERS
Data Science	6,327
Information Technology	4,624
Computer Science	4,353
Personal Development	3,221
Language Learning	2,844

Catalogue average	**3,212**

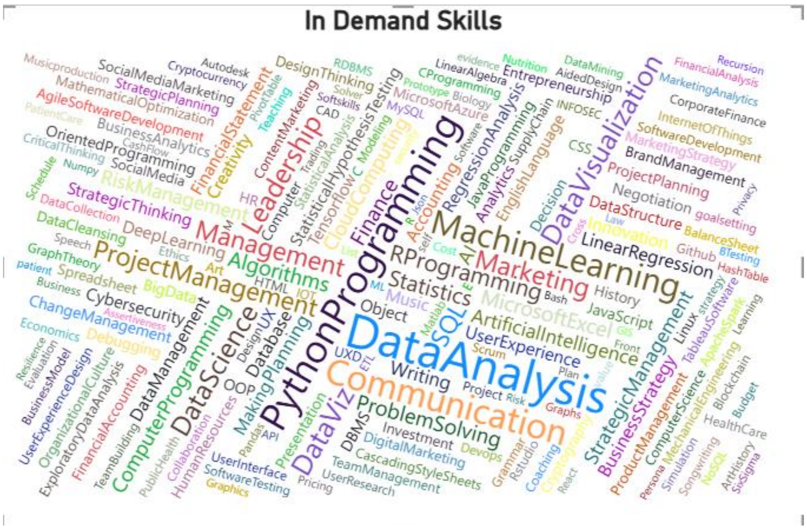
Personal Development and Language Learning sit at the bottom. The language split is stark: within Data Science, English averages 6,364 views against 1,740 for Spanish and 230 for Japanese. Information Technology is sharper still — 4,886 for English, 31 for Japanese.

One exception stands out. In Language Learning, French courses average 4,468 views against 2,825 for English — the only place in the entire catalogue where a non-English language outperforms English. It is a narrow signal, but a genuine one.

> **WHAT THIS MEANS**
> Technical categories pull roughly double the engagement of soft-skill ones. Data Science, IT, and Computer Science are the priority areas for investment. The French result in Language Learning is the one case where non-English content demonstrably outperforms, and is worth treating as a distinct opportunity rather than an anomaly to ignore.

FINDING 03

Most commonly taught skills



In-demand skills across the catalogue

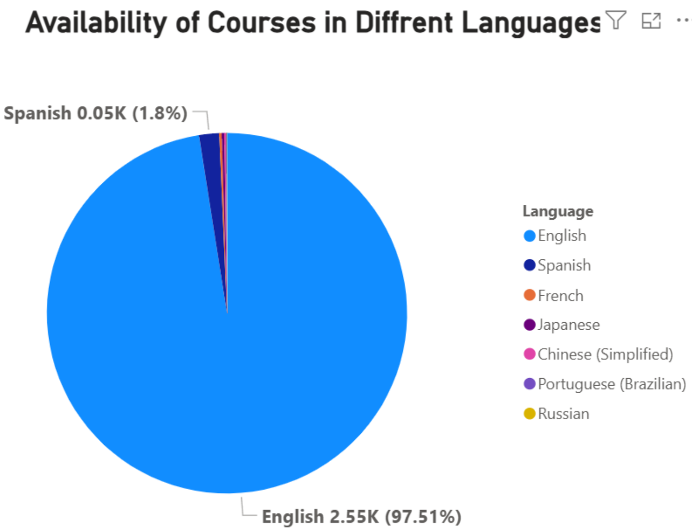
The catalogue covers **13 distinct skills**. The most common are Data Analysis, Python, Communication, and Machine Learning — technical and quantitative, with Communication the only soft skill in the top tier. **Python Programming** is the most cross-cutting skill, leading in all three of the highest-volume categories.

13	**#1**
Distinct skills taught	Python, most cross-cutting

> **WHAT THIS MEANS**
> Python has become general-purpose infrastructure rather than a specialist skill — a low-risk content investment because learners arrive at it from several directions. The narrow overall skill set also suggests room to differentiate on skills the market underserves.

FINDING 04

Language distribution



Course availability by language

Courses exist in seven languages: English, Spanish, French, Japanese, Russian, Chinese, and Portuguese.

97.5%	1.8%	0.23%	<0.5%
English	Spanish	French	Remaining four

WHAT THIS MEANS

The platform is effectively monolingual. This is a *supply* observation, not a *demand* one — there's so little non-English content that low non-English viewership can't tell us whether the audience exists.

FINDING 05

Language preference in the top 5 categories

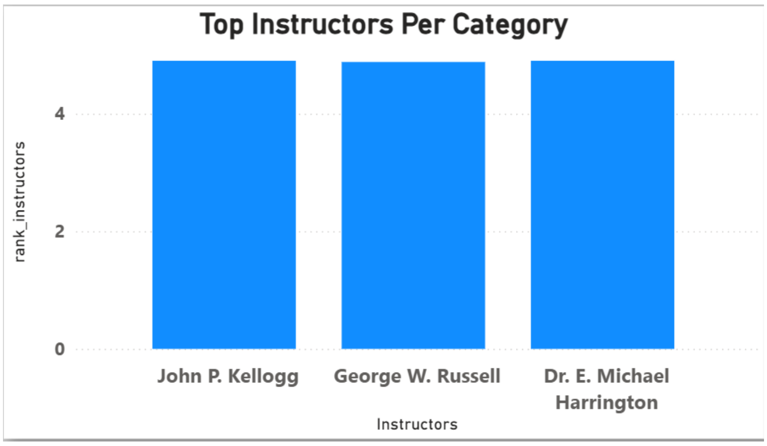
English leads in all five, with no category close to breaking the pattern. The one signal worth noting: **Marketing** carries a noticeably larger Spanish share than anywhere else in the catalogue — still well below English, but visible.

WHAT THIS MEANS

Spanish-language Marketing content is worth a small pilot to test whether demand exists. A broader localisation programme isn't supported by anything in this data.

FINDING 06

Top instructors by category



Highest-rated instructors, static ranking

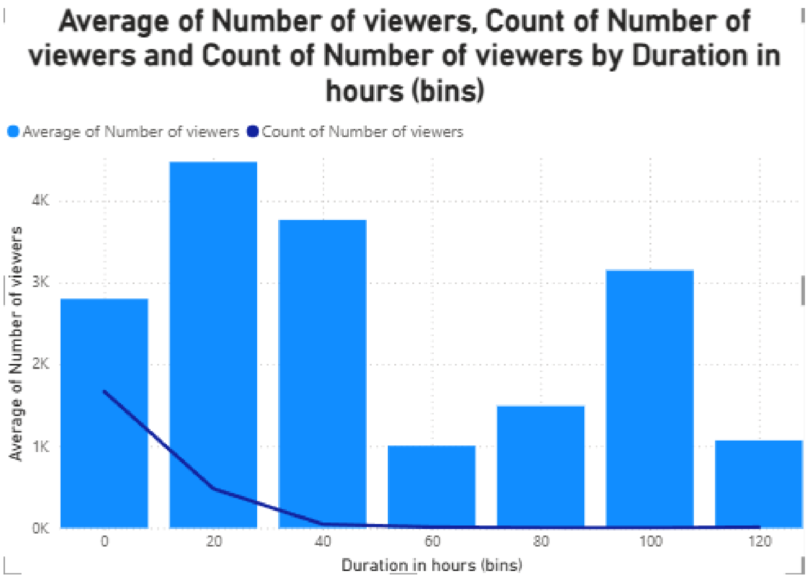
Ranked on average rating, the leading instructors are John P. Kellogg, George W. Russell, and Dr. E. Michael Harrington, each at or near a perfect 5.0. This visual is static — it doesn't respond to slicers elsewhere on the page.

> **WHAT THIS MEANS**
>
> These are the client's strongest partnership candidates. But rating alone is a fragile criterion — three instructors tied at a perfect 5.0 suggests small review volumes rather than genuine separation. A better shortlist would weight rating by review count, so an instructor with 5.0 from ten reviews doesn't outrank one with 4.8 from a thousand.

FINDING 07

Course duration and viewership

This one needed care. The initial chart plotted average views against every individual duration value and showed dramatic spikes, one reaching 26,000 views around 650 hours. Adding a course count alongside the average revealed why: almost the entire catalogue sits below 100 hours, and course counts collapse to single digits beyond that. Those spikes were individual courses, not trends.



Average and count of viewers by duration bins, filtered to courses under 150 hours

Restricting to courses under 150 hours and binning into 20-hour bands gives a reliable picture:

20–40h	1k–3.2k	60.32h
Peak engagement band (~4,400 avg views)	Views beyond peak, no clear trend	Catalogue average duration

Average duration by category is fairly uniform, from 41.96 hours (Personal Development) to 67.82 hours (Math & Logic).

WHAT THIS MEANS

There's a modest sweet spot around 20–40 hours, but duration isn't a strong engagement lever. Set course length by subject matter, not in pursuit of views. The apparent advantage of very long courses in the raw data is a sampling artifact.

FINDING 08

Skill variety and viewership

Category	Average of No of Skills	Average of Duration in hours
Data Science	4.71	63.99
Personal Development	4.13	41.96
Arts and Humanities	4.02	60.03
Computer Science	4.00	65.29
Information Technology	3.77	54.73
Business	3.77	59.41
Social Sciences	3.22	49.13
Health	2.94	55.72
Physical Science and Engineering	2.80	63.01
Math and Logic	2.59	67.82
Language Learning	1.81	66.82
Total	**3.75**	**60.32**

Average skills and duration by category

Average skill count per category runs from **1.81** (Language Learning) to **4.71** (Data Science), against a catalogue average of 3.75. Comparing this against viewership shows no meaningful relationship:

- Personal Development — 2nd highest skill variety (4.13), near-bottom on views
- Information Technology — below-average variety (3.77), 2nd highest on views
- Data Science leads on both, but the pattern breaks immediately below it

All courses in the dataset are recorded online courses, so no additional filter was needed here.

WHAT THIS MEANS

Packing more skills into a course doesn't drive viewership. Subject matter matters far more than breadth. The client should prioritise topic selection over skill count.

Note

This project began as a tutorial build and was extended with independent analysis, including the duration investigation in Finding 07 and the reasoning behind each implication.